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 Review the assignment due date

Final Project

Team Number: T14

Team Name: doomsDAY

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GitHub Repository URL: <https://github.com/upenn-embedded/final-project-s26-t14.git>

GitHub Pages Website URL: <https://upenn-embedded.github.io/final-project-s26-t14/>

Final Project Proposal

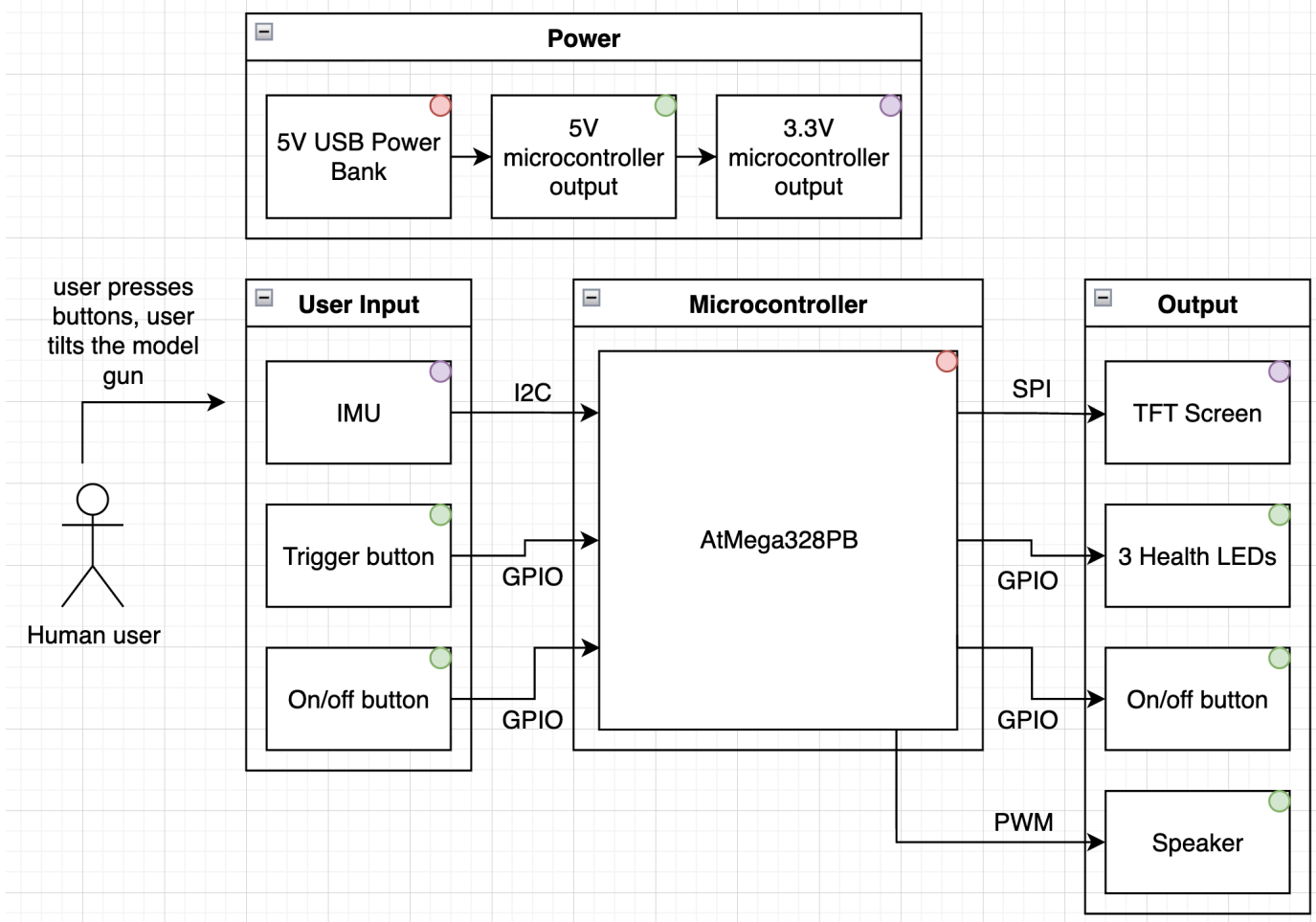
1. Abstract

The Mini Arcade Station features a simplified smash-TV shooting game with an external model turret gun in place of joy sticks. It has a TFT screen to display menu and game, signal lights, a buzzer/speaker to generate sound output, and an IMU sensor to determine turret angle.

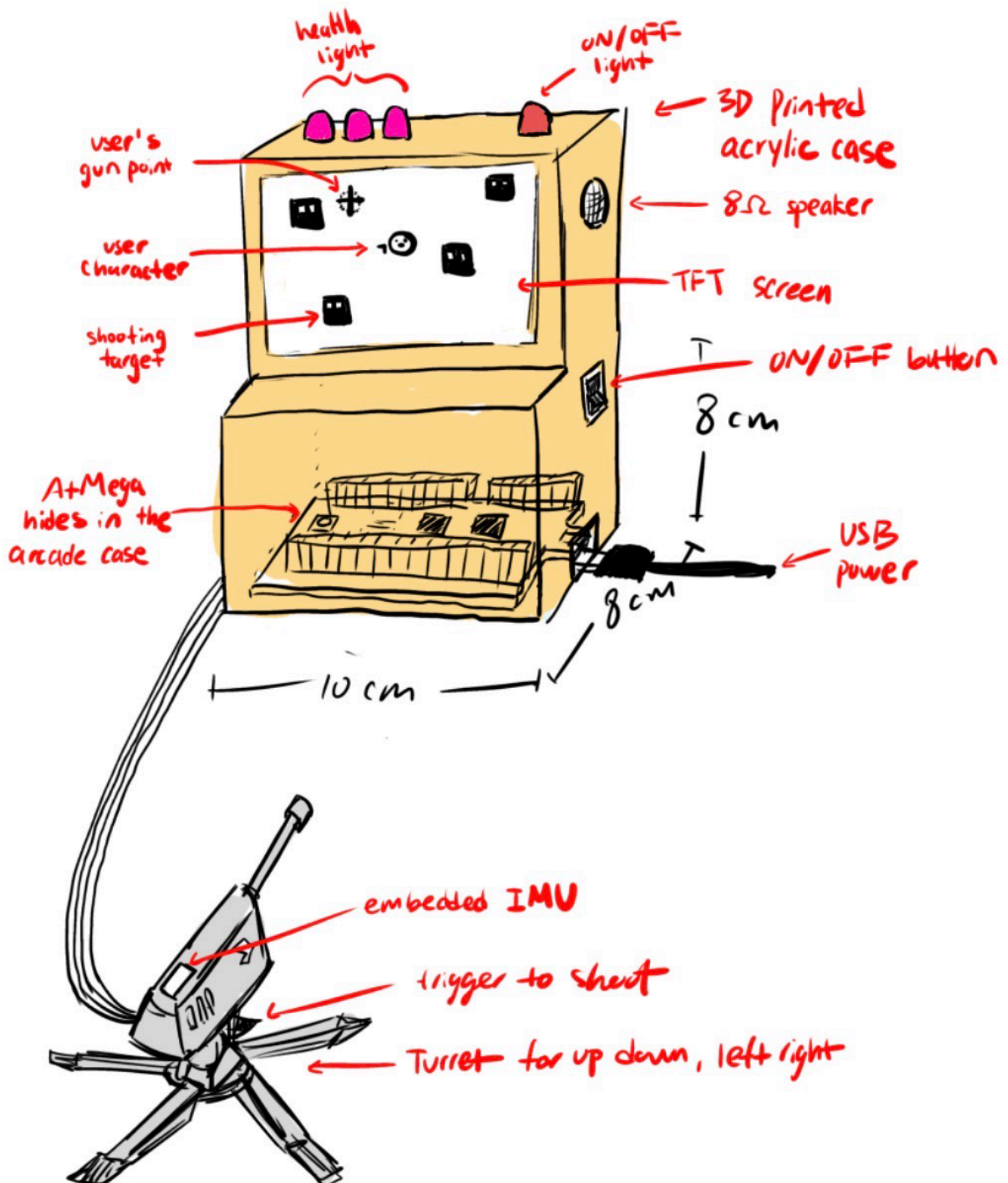
2. Motivation

For retro arcade lovers and parents on a tight budget, the Mini Arcade Station is perfect to have while occupying a small space. This project not only allows gamer folks to play an arcade-style game within the homes, but it also takes a child's attention away from the phone and onto a physical game that is tactile and interactive. The shooting game itself is inspired from Stardew Valley's Journey of the Prairie King, a minigame which Amaris enjoys.

3. System Block Diagram



4. Design Sketches



5. Software Requirements Specification (SRS)

5.1 Definitions, Abbreviations

GPIO: General Purpose Input Output

LED: Light Emitting Diode

I2C: Inter-Integrated Circuit

SPI: Serial Peripheral Interface

TFT LCD: Thin-Film Transistor Liquid Crystal Display

5.2 Functionality

ID	Description
SRS-01	The IMU 3-axis acceleration will be used to measure the position of the gun and where the cursor should be on screen relative to gun position.
SRS-02	Use interrupts to handle button presses and game states (such as reloading, shooting, killing enemies, etc.)
SRS-03	I2C communication will be used to communication data between the gun and the main arcade machine.
SRS-04	The health of the player should be displayed through the LED indicators at the top of the arcade machine.
SRS-05	The buzzer should play sounds when an action is performed (ie. shooting).
SRS-06	The score of the player should be displayed.

6. Hardware Requirements Specification (HRS)

6.1 Definitions, Abbreviations

GPIO: General Purpose Input Output

LED: Light Emitting Diode

I2C: Inter-Integrated Circuit

SPI: Serial Peripheral Interface

TFT LCD: Thin-Film Transistor Liquid Crystal Display

6.2 Functionality

ID	Description
HRS-01	ATmega328PB is the main microcontroller for this design.
HRS-02	The TFT display will be a 3.5" SPI-interfaced color LCD with a minimum resolution of 320x480 pixels and support for 16-bit color depth.
HRS-03	The MPU-6050 IMU shall be powered at 3.3V and communicate with the gun ATmega328PB over I2C.
HRS-04	The trigger button shall be connected to external interrupt pin with internal pull-up resistor, producing falling edge signal detections.
HRS-05	The speaker shall produce audible output at frequencies for sound effects, and game start & end indicator.
HRS-06	Each LED shall illuminate when driven by GPIO pins through current limiting resistors.
HRS-07	A on/off switch will be used to turn the arcade machine on/off.

7. Bill of Materials (BOM)

[Link to BOM](#)

TFT Display: Primary output display. SPI interface satisfies serial peripheral requirement. 320x480 provides adequate sprite resolution.

IMU: 6-axis gyro + accelerometer. I2C interface. Used for 4-direction aim detection in gun prop.

Speaker: Audio feedback on shot events.

LED: 3 red LEDs for health bar, 1 green LED for power indicator. GPIO-driven.

Button: Trigger button in gun prop.

Switch: Turns machine on/off.

Power Bank: Safe, portable power. Avoids LiPo battery restriction. Sufficient capacity for full demo session.

8. Final Demo Goals

The demo is fully self-contained and requires only a standard table. No outdoor space, mounting infrastructure, or special environment is needed.

Demo Flow

1. Power on the cabinet via the USB power bank. The power LED illuminates green and the display shows the start screen.
2. Pick up the gun prop. The game graphics appears on the TFT screen.
3. Zombies begin spawning from the screen edges and advancing toward the center.
4. Demonstrate aiming: tilt the gun up, down, left, and right. The crosshair moves accordingly.
5. Pull the trigger to shoot a zombie. The speaker buzzes, the zombie is removed, and the score increments.
6. Allow one zombie to approach the player. One health LED turns off.
7. Allow all 3 zombies to pass. All LEDs extinguish and the Game Over screen appears.

9. Sprint Planning

Milestone	Functionality Achieved	Distribution of Work
Sprint #1	- Fully implement electronics hardware, wire everything, 3D print turret. - Begin implementing basic inputs & outputs (speaker, LED, buttons)	Hardware - All 3D print - Daniel & Yi Lu
Sprint #2	- Connecting TFT screen to MCU, attaching IMU to turret and MCU - Program starting screen and user sprite to TFT	TFT - Yi Lu IMU - Daniel Game - Amaris
MVP Demo	- Demonstrate button trigger outputs speaker sound, moving turret moves user crosshair	All
Final Demo	- Demonstrate working game (spawn player and zombies) - Demonstrate laser cut casing with decorations	All Laser cutting - Daniel & Yi Lu

This is the end of the Project Proposal section. The remaining sections will be filled out based on the milestone schedule.

Sprint Review #1

Last week's progress

Current state of project

Next week's plan

Sprint Review #2

Last week's progress

Current state of project

Next week's plan

MVP Demo

Final Report

Don't forget to make the GitHub pages public website! If you've never made a GitHub pages website before, you can follow this webpage (though, substitute your final project repository for the GitHub username one in the quickstart guide):

<https://docs.github.com/en/pages/quickstart>

1. Video

2. Images

3. Results

3.1 Software Requirements Specification (SRS) Results

ID	Description	Validation Outcome
SRS-01	The IMU 3-axis acceleration will be measured with 16-bit depth every 100 milliseconds +/-10 milliseconds.	Confirmed, logged output from the MCU is saved to "validation" folder in GitHub repository.

3.2 Hardware Requirements Specification (HRS) Results

ID	Description	Validation Outcome
HRS-01	A distance sensor shall be used for obstacle detection. The sensor shall detect obstacles at a maximum distance of at least 10 cm.	Confirmed, sensed obstacles up to 15cm. Video in "validation" folder, shows tape measure and logged output to terminal.

4. Conclusion

References
